APPENDIX II—UPDATED DEBT SUSTAINABILITY ANALYSIS[1]

Ghana's risk of debt distress is assessed to remain moderate, which is in line with the 2009 debt sustainability analysis (DSA). Public sector debt dynamics are broadly sustainable in light of the scale and evolution of the domestic debt stock, but suggest that even a stronger fiscal effort might be desirable than suggested in the baseline. However, the external and public DSAs suggest that vulnerabilities have increased compared to the previous DSA, reflecting the projected higher and more front-loaded nonconcessional borrowing. A low growth outcome associated with high investments substantially increases the risks to public debt sustainability.

I. BACKGROUND

1. **The 2009 debt sustainability analysis pointed to improvements relative to the previous year in the debt sustainability baseline while also highlighting short- and medium-term vulnerabilities.** Despite the favorable baseline trend, risk of debt distress was considered to be moderate. The risks to the baseline scenario were concentrated on the government's ability to sustain fiscal consolidation over the medium-to-long term and to successfully increase the growth potential of the nonoil sector through investing oil and gas revenues in a productive matter. A moderately higher recourse to external nonconcessional borrowing by the public sector over the medium term, reflecting Ghana's emerging middle-income status, was viewed as sustainable.

2. **The public debt burden at end-2009 was below the level projected in the 2009 DSA (Text Table 1).** Outstanding public debt and total external debt was 10 percentage points below the projection in last year's DSA, reflecting a confluence of factors. First, the cedi exchange rate strengthened against the U.S. dollar from mid-2009, ending the year at a more appreciated level than projected in the 2009 DSA. This exchange rate outcome reduced the public debt-to- GDP ratio in 2009 by more than 4 percentage points of GDP relative to the 2009 DSA projection. Second, loans from the World Bank ($150 million) and from the IMF under the ECF arrangement ($100 million) anticipated in the 2009 DSA to be disbursed at end-2009 are now projected to be disbursed in mid-2010 (these combined loans were equivalent to 1.6 percent of 2009 GDP). Third, there was less commercial borrowing than anticipated in 2009, see Text Table 2, consistent with the much smaller current account deficit than earlier projected.

[1] This is an update to last year's DSA (IMF Country Report No. 09/256) that was included in the original ECF (PRGF) program request for Ghana.

Text Table 1: Revised Public Debt Situation

	2009 DSA			2010 DSA		
	2008 Est.	2009 Proj.	2010 Proj.	2008 Act.	2009 Est.	2010 Proj.
	(in percent of GDP)					
External debt [1]	42.8	57.2	63.4	41.0	47.1	51.3
Public debt	58.2	70.0	72.6	57.6	60.4	68.4

Sources: Ghanaian authorities and staff estimates and projections.

[1] Public and private sector external debt.

II. MACROECONOMIC ASSUMPTIONS FOR THE DSA

3. **Compared to the 2009 DSA, macroeconomic assumptions have been modified to reflect more recent data (Text Table 2).** While the 2009–10 output-inflation performance is less favorable, projections for the medium-to-long term are broadly unchanged. The latest staff estimates of real growth in 2009 are one percentage point below the previous projection and reflect the effects of the global financial crisis and tighter domestic policies, with a smaller shortfall in 2010. Inflation was about three percentage points higher than previously projected for 2009, with the gap narrowing to less than two percentage points for 2010. In 2009, the current account deficit also narrowed more than projected in the 2009 DSA, with a beneficial effect on external financing needs. While the cash-based primary fiscal balance in 2009 improved relative to the 2009 DSA, if measured on commitment-basis, the primary deficit was much higher because of the accumulation of new domestic payment arrears totaling 4.2 percent of GDP.

4. **The current DSA assumes more front-loaded nonconcessional borrowing.** This reflects the new framework on external debt limits for low-income countries, which would provide more flexibility in Ghana's nonconcessional external borrowing (Text Table 2).[2] The more notable difference is for the period 2010–15, where the 2010 DSA assumes an increase in nonconcessional external borrowing to 4.5 percent of GDP on an annual average basis, up from a projected 2.4 percent of GDP in the 2009 DSA[3].

[2] Ghana was classified as a country with low debt vulnerabilities and low capacity. The low debt vulnerability was established based on the DSA (IMF Country Report No. 09/256), while the low capacity based on the PEFA and CPIA indices and after discussing the quality of macroeconomic public financial management institutions jointly by the Bank and the Fund.

[3] The projected new external nonconcessional borrowing has an average interest of 8 percent and average maturity of 8 years (with 2 years grace).

Text Table 2: Key Macroeconomic Assumptions

	2009 DSA				2010 DSA			
	2009 Proj.	2010 Proj.	Average 2010-15 Proj.	Average 2016-29 Proj.	2009 Act.	2010 Proj.	Average 2010-15 Proj.	Average 2016-30 Proj.
Real GDP growth [1]	4.5	5.0	8.6	4.7	3.5	4.5	7.8	4.9
Inflation (GDP deflator) [1]	17.2	11.0	6.7	6.2	20.7	12.8	7.3	6.1
Current account deficit [2]	-13.0	-15.9	-9.2	-7.8	-5.1	-12.1	-7.7	-6.5
Nonconcessional external borrowing [2]	3.4	3.2	2.4	5.6	0.6	3.6	4.5	5.0
Primary fiscal balance [3]	-5.3	-2.4	-0.2	0.1	-5.0	-2.9	0.3	0.0

Sources: Ghanaian authorities and staff estimates and projections.

[1] In annual percentage change.

[2] In percent of GDP.

[3] In percent of GDP and on cash basis.

III. EXTERNAL DSA

5. **With the assumed level of external nonconcessional borrowing, risks of debt distress have risen but appear manageable (Table 1).** The external DSA shows that the thresholds on the present values of debt-to-GDP, debt-to-exports and debt-to-revenue ratios are not violated, though the baseline scenario gets close to the debt-to-GDP ratio threshold between 2014 and 2019. The thresholds on the debt service-to-exports and debt-service-to-revenue ratios are not violated either, but the 2007 sovereign Euro bond redemption in 2017 is associated with a marked spike in debt service indicators.

6. **Standard stress tests underlie the importance of fiscal consolidation as Ghana enters the oil era (Figure 1).** Under an alternative scenario based on historical growth projections, debt burden indicators rise rapidly over the projection period, breaching most indicative thresholds. This, however, fails to account for the soon-to-begin oil production, and hence understates the long-term growth potential and overstates the fiscal financing needs. Under bound tests,[4] the most extreme stress test points to breaches of some of the indicative thresholds. The stress tests underscore the risks related to potential delays in oil production.

[4] The standardized bound tests have been modified so that the shock influences years 2012–13, to account for the fact that real growth and exports are both projected to surge with the start of oil production in 2011 (20 and 37 percent, respectively). Using historic average growth rates minus one standard deviation therefore represent a dramatic revision to baseline projection for 2011 since this effectively eliminates the benefits of the oil sector from the DSA baseline without reducing imports financed by oil revenues.

IV. PUBLIC SECTOR DSA

7. **Ghana's fiscal stance has a critical bearing on public debt sustainability in the projection period (Table 2).** This highlights the importance of an early consolidation that would lead to much lower debt burden. The baseline scenario, which assumes a strong fiscal consolidation, also points to noticeable uncertainty related to the budget's capacity to service debt. Even under this strong fiscal adjustment, the debt service-to-revenue ratio is projected to stabilize around 35–40 percent over the medium term, after increasing in 2017 with the redemption of the 2007 Euro bond. A further risk to public debt sustainability arises from the recapitalization of two state-owned banks and from the external debt of the five SOEs (see MEFP ¶41) that will be included in the definition of public and publicly guaranteed external debt in the future.

8. **Public debt dynamics are sensitive to growth and fiscal consolidation (Figure 2)**. The consequences of a delay in fiscal adjustment or a temporary shock to growth are well reflected in the bound tests. In both cases, the risks of debt distress become high.

9. **A sustained low growth outcome would further increase the risks to public debt sustainability.** This is illustrated by a scenario where high public investment fails to generate the sustained high long-term growth assumed in the baseline scenario. Under an alternative low growth scenario,[5] a substantial fiscal deterioration over the medium- to long-term causes a rapid build-up of public debt. Risks of debt distress are higher than under the bound tests.

V. CONCLUSIONS

10. **Vulnerabilities have increased compared to last year's exercise.** The short-term trajectories have been influenced by the more pronounced economic slowdown and weaker fiscal adjustment effort during 2009–10. Over the medium to long term, the picture points to rising debt vulnerabilities (Text Tables 3 and 4). The vulnerabilities largely reflect the more front-loaded external borrowing at nonconcessional terms by the public sector assumed in the 2010 DSA exercise compared to the previous DSA, which has increased the debt levels and debt-service ratios in the medium-term. However, the debt vulnerability indicators stabilize in the long-run and continue to stay below their respective threshold levels in the baseline scenario, as in the 2009 DSA. Therefore, we maintain the assessment of **moderate risk of debt distress**. The analysis suggests that rapidly rising external borrowing at commercial terms would lead to increasing risks to debt sustainability. The public sector debt dynamics appear sustainable in light of the current size and the evolution of the

[5] In the low growth scenario, annual real growth during 2012–30 is lowered from the average of about 5 percent in the baseline scenario to 2.6 percent.

domestic debt stock, but point to the need for strong fiscal consolidation. The baseline has increased, largely on account of increased external debt. Moreover, a weak growth response to higher investment has significant adverse implications for public debt sustainability.

Text Table 3: Indicators of External Debt Vulnerability (Baseline)

	2015	2020
PV of debt-to -GDP ratio		
2009 DSA	29.5	30.7
2010 DSA	42.9	38.5
Threshold	*50*	*50*
PV of debt-to-exports ratio		
2009 DSA	51.0	60.0
2010 DSA	104.5	107.2
Threshold	*200*	*200*
PV of debt-to-revenue ratio		
2009 DSA	109.8	110.8
2010 DSA	150.6	144.5
Threshold	*300*	*300*
Debt service-to-exports ratio		
2009 DSA	6.7	9.6
2010 DSA	10.1	17.8
Threshold	*25*	*25*
Debt service-to-revenue ratio		
2009 DSA	14.5	17.6
2010 DSA	14.5	24.0
Threshold	*35*	*35*

Text Table 4: Indicators of Public Debt Vulnerability (Baseline)

	2015	2020
PV of debt-to -GDP ratio		
2009 DSA	39.9	35.9
2010 DSA	60.3	48.6
PV of debt-to-revenue ratio		
2009 DSA	133.3	121.5
2010 DSA	195.4	174.2
Debt service-to-revenue ratio		
2009 DSA	20.8	24.6
2010 DSA	22.0	33.8

Table 1: External Debt Sustainability Framework, Baseline Scenario, 2007-30 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical 0 Average	Standard 0 Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	39.2	41.0	47.1	86.7	47.8	51.3	46.2	48.3	52.6	55.2	56.9	51.8	50.9	36.3	46.5
o/w public and publicly guaranteed (PPG)	24.8	27.8	32.5	71.4	47.0	38.2	35.8	38.9	43.6	46.4	48.3	41.9	43.2	31.2	39.6
Change in external debt	5.9	1.8	6.1	-7.6	25.4	4.2	-5.1	2.1	4.3	2.6	1.6	1.6	-1.5	-1.4	-1.4
Identified net debt-creating flows	1.3	7.6	0.5	0.9	27.5	-0.8	-9.9	-3.7	-3.8	-4.4	-3.9	-4.4	0.7	-1.0	-0.6
Non-interest current account deficit	11.7	17.8	4.2	6.0	5.7	10.4	6.7	6.0	5.9	2.9	3.2	5.9	6.9	2.3	5.0
Deficit in balance of goods and services	27.0	33.0	19.2	21.4	6.3	24.4	13.9	12.5	13.3	11.8	13.0	14.8	20.0	15.9	18.2
Exports	40.0	42.5	50.3	42.6	4.4	48.7	51.0	45.5	43.4	42.1	41.0	45.3	35.9	34.6	35.9
Imports	67.0	75.5	69.6	64.0	6.6	73.1	64.9	57.9	56.7	53.9	54.0	60.1	55.9	50.5	54.1
Net current transfers (negative = inflow)	-15.9	-15.9	-16.0	-15.9	2.0	-14.3	-13.2	-12.5	-13.0	-14.0	-14.3	-13.5	-13.7	-13.2	-13.6
o/w official	-3.7	-4.1	-4.5	-3.9	0.8	-4.3	-3.2	-2.0	-2.0	-2.0	-1.8	-2.5	-0.7	-0.2	-0.6
Other current account flows (negative = net inflow)	0.6	0.7	1.1	0.5	0.3	0.3	6.1	5.9	5.5	5.2	4.5	4.6	0.6	-0.4	0.4
Net FDI (negative = inflow)	-5.7	-7.3	-7.7	-3.2	2.8	-11.1	-10.2	-8.5	-9.3	-7.0	-7.0	-8.8	-6.0	-3.0	-5.4
Endogenous debt dynamics 2/	-4.8	-3.0	4.0	-1.9	26.4	-0.1	-6.5	-1.2	-0.5	-0.3	-0.2	-1.5	-0.2	-0.3	-0.2
Contribution from nominal interest rate	0.3	0.8	0.9	1.6	0.9	1.8	1.4	1.6	1.9	2.1	2.3	1.8	2.0	1.7	1.9
Contribution from real GDP growth	-1.6	-2.6	-1.5	-4.4	2.0	-1.8	-7.9	-2.7	-2.4	-2.5	-2.5	-3.3	-2.2	-1.9	-2.1
Contribution from price and exchange rate changes	-3.5	-1.3	4.6	0.9	26.7	…	…	…	…	…	…	…	…	…	…
Residual (3-4) 3/	4.7	-5.8	5.5	-8.5	17.2	5.1	4.8	5.8	8.2	7.0	5.6	6.1	-2.2	-0.5	-0.8
o/w exceptional financing	0.0	0.0	-0.3	-2.3	2.6	-1.5	0.0	0.0	0.0	0.0	0.0	-0.3	0.0	0.0	0.0
PV of external debt 4/	...	...	42.1	42.1	…	45.0	40.8	43.1	47.4	50.0	51.5	46.3	46.2	35.1	42.9
In percent of exports	...	...	83.6	83.6	…	92.6	79.9	94.9	109.2	118.6	125.6	103.5	128.8	101.4	119.2
PV of PPG external debt	...	...	27.5	27.5	…	32.0	30.4	33.8	38.3	41.2	42.9	36.4	38.5	30.0	36.0
In percent of exports	...	...	54.6	54.6	…	65.7	59.6	74.3	88.4	97.8	104.7	81.7	107.3	86.7	100.1
In percent of government revenues	...	...	122.3	122.3	…	123.3	117.3	130.6	148.9	149.2	150.8	136.7	144.7	118.7	136.5
Debt service-to-exports ratio (in percent)	9.8	10.8	9.8	19.4	7.1	10.2	7.6	8.5	9.4	11.7	14.1	10.2	21.9	18.3	20.5
PPG debt service-to-exports ratio (in percent)	3.0	4.6	4.2	12.9	6.9	5.0	3.6	4.5	5.3	7.6	10.1	6.0	17.8	15.5	16.9
PPG debt service-to-revenue ratio (in percent)	5.3	8.6	9.4	28.1	18.9	9.4	7.1	7.9	9.0	11.6	14.6	9.9	24.0	21.2	23.0
Total gross financing need (Millions of U.S. dollars)	1494	2522	237	1220	743	751	99	361	199	265	667	390	3901	5834	4145
Non-interest current account deficit that stabilizes debt ratio	5.8	16.1	-1.8	13.6	25.0	6.1	11.8	3.8	1.6	0.3	1.6	4.2	8.4	3.7	6.4
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.7	7.3	3.5	5.2	1.2	4.5	20.1	6.8	5.3	5.0	4.8	7.8	4.5	5.7	4.9
GDP deflator in US dollar terms (change in percent)	11.6	3.4	-10.0	3.4	16.4	10.3	9.0	7.6	2.5	1.6	1.6	5.4	2.4	4.1	2.7
Effective interest rate (percent) 5/	1.1	2.4	2.1	1.8	0.4	4.3	3.6	3.9	4.2	4.3	4.5	4.1	4.1	4.9	4.4
Growth of exports of G&S (US dollar terms, in percent)	17.5	17.8	10.4	12.6	9.6	11.5	37.3	2.4	3.0	3.5	3.7	10.2	5.2	9.2	6.5
Growth of imports of G&S (US dollar terms, in percent)	21.5	24.9	-14.1	12.1	16.9	21.1	16.3	2.6	5.6	1.3	6.7	8.9	7.1	7.7	7.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	13.9	4.2	-0.6	-3.5	-5.4	-6.2	0.4	-8.3	-13.1	-9.9
Government revenues (excluding grants, in percent of GDP)	22.7	22.8	22.5	20.8	2.2	25.9	25.9	25.9	25.7	27.6	28.5	26.6	26.6	25.2	26.3
Aid flows (in Millions of US dollars) 7/	917	776	776	525	278	1365	1134	951	1006	1052	1033	1090	814	641	742
o/w Grants	917	776	776	525	278	921	922	701	756	802	783	814	564	391	492
o/w Concessional loans	0.3	0.2	0.2	0.2	0.1	444	212	250	250	250	250	276	250	250	250
Grant-equivalent financing (in percent of GDP) 8/	...	...	...	…	…	6.6	4.3	2.6	2.3	2.2	2.0	3.3	0.8	-0.3	0.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...	…	…	42.2	37.3	23.6	23.9	24.4	22.5	29.0	10.5	-5.6	5.1
Memorandum items:															
Nominal GDP (Millions of US dollars)	15018	16654	15513	10363	4412	17886	23415	26915	29035	30970	32959	26863	44526	100389	59894
Nominal dollar GDP growth	17.9	10.9	-6.9	8.9	17.8	15.3	30.9	14.9	7.9	6.7	6.4	13.7	7.0	10.0	7.7
PV of PPG external debt (in Millions of US dollars)			4235	…	…	5657	7120	9090	10947	12542	13921	9880	16866	29593	20452
Gross remittances (Millions of US dollars)	…	…	…	…	…	…	…	…	…	…	…		…	…	
PV of PPG external debt (in percent of GDP + remittances)	...	...	27.5	…	…	32.0	30.4	33.8	38.3	41.2	42.9	…	38.5	30.0	36.0
PV of PPG external debt (in percent of exports + remittances)	...	...	54.6	…	…	65.7	59.6	74.3	88.4	97.8	104.7	…	107.3	86.7	100.1
Debt service of PPG external debt (in percent of exports + remittances)	...	...	4.2	…	…	5.0	3.6	4.5	5.3	7.6	10.1	…	17.8	15.5	16.9

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

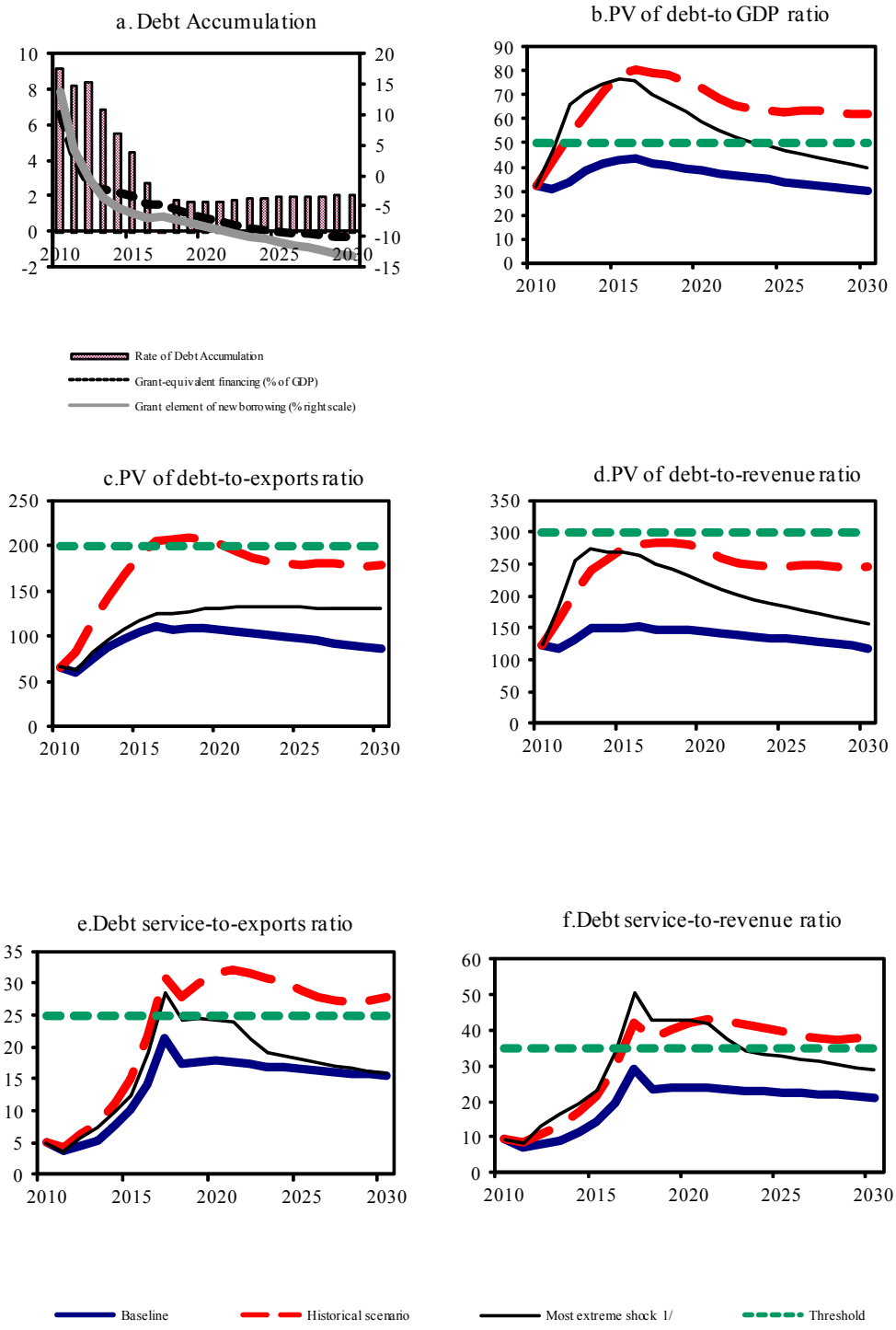
5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure 1: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-30 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Combination shock; in c. to a Terms shock; in d. to a Combination shock; in e. to a Non-debt flows shock and in figure f. to a Combination shock

Table 2: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-30
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	51.2	57.6	60.2	96.2	46.8	68.2	61.1	63.2	67.4	67.3	65.7	65.5	53.3	43.0	50.2
o/w foreign-currency denominated	24.8	27.8	32.5	71.4	47.0	38.2	35.8	38.9	43.6	46.4	48.3	41.9	43.2	31.2	39.6
Change in public sector debt	9.1	6.4	2.6	-8.2	23.9	8.0	-7.1	2.1	4.2	0.0	-1.7	0.9	-1.6	0.0	-1.5
Identified debt-creating flows	0.8	2.8	1.5	-3.9	26.5	1.5	-12.3	-4.4	-0.5	-3.6	-3.9	-3.9	-1.2	-0.7	-1.4
Primary deficit	6.1	10.7	5.0	3.2	3.4	2.9	-0.1	-0.2	0.3	-2.2	-2.8	-0.3	0.1	1.2	0.0
Revenue and grants	28.8	27.5	27.5	25.6	3.4	31.1	29.9	28.5	28.4	30.2	30.8	29.8	27.9	25.6	27.3
of which: grants	6.1	4.7	5.0	4.9	1.6	5.1	3.9	2.6	2.6	2.6	2.4	3.2	1.3	0.4	1.0
Primary (noninterest) expenditure	34.9	38.1	32.5	28.8	5.5	34.0	29.8	28.2	28.6	28.0	28.1	29.4	28.0	26.8	27.3
Automatic debt dynamics	-3.8	-1.7	-3.2	-6.0	25.7	-2.7	-11.9	-3.9	-0.5	-1.2	-0.9	-3.5	-1.2	-1.8	-1.3
Contribution from interest rate/growth differential	-2.9	-3.9	-2.8	-5.5	1.9	0.1	-9.3	-2.1	-0.9	-1.3	-1.1	-2.4	-1.0	-1.2	-1.0
of which: contribution from average real interest rate	-0.7	-0.5	-0.9	-0.6	0.6	2.7	2.1	1.8	2.3	1.9	2.0	2.1	1.4	1.1	1.4
of which: contribution from real GDP growth	-2.3	-3.5	-2.0	-4.9	1.9	-2.6	-11.4	-3.9	-3.2	-3.2	-3.1	-4.6	-2.4	-2.3	-2.4
Contribution from real exchange rate depreciation	-0.8	2.2	-0.4	-0.5	25.2	-2.7	-2.6	-1.8	0.4	0.1	0.2	-1.1	...	...	-0.1
Other identified debt-creating flows	-1.5	-6.1	-0.3	-1.1	1.9	1.2	-0.3	-0.3	-0.3	-0.2	-0.2	0.0	-0.1	0.0	-0.1
Privatization receipts (negative)	-0.8	-5.7	0.0	-0.7	1.8	1.7	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	-0.7	-0.4	-0.3	-0.5	0.5	-0.5	-0.3	-0.3	-0.3	-0.2	-0.2	-0.3	-0.1	0.0	-0.1
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residual, including asset changes	8.3	3.6	1.1	-4.3	13.3	6.5	5.2	6.5	4.7	3.5	2.2	4.8	-0.4	0.7	-0.2
Other Sustainability Indicators															
PV of public sector debt	26.4	29.7	55.2	27.7	11.1	62.0	55.6	58.0	62.1	62.1	60.3	60.0	48.6	41.8	46.6
o/w foreign-currency denominated	0.0	0.0	27.5	3.1	9.2	32.0	30.4	33.8	38.3	41.2	42.9	36.4	38.5	30.0	36.0
o/w external	...	...	27.5	…	…	32.0	30.4	33.8	38.3	41.2	42.9	36.4	38.5	30.0	36.0
PV of contingent liabilities (not included in public sector debt)	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Gross financing need 2/	16.8	22.5	18.7	18.4	3.6	17.1	12.7	11.5	11.5	9.3	9.2	11.9	14.2	15.3	14.1
PV of public sector debt-to-revenue and grants ratio (in percent)	91.5	108.3	200.8	95.9	52.4	199.4	186.3	203.8	219.1	205.6	195.6	201.6	174.4	163.0	170.5
PV of public sector debt-to-revenue ratio (in percent)	116.0	130.4	245.5	118.5	62.8	239.0	214.6	224.3	241.2	224.9	211.9	226.0	182.6	165.5	176.9
o/w external 3/	…	…	122.3	122.3		123.3	117.3	130.6	148.9	149.2	150.8	136.7	144.7	118.7	136.5
Debt service-to-revenue and grants ratio (in percent) 4/	19.3	24.0	28.1	42.6	22.0	26.9	23.6	21.8	20.2	20.1	22.0	22.4	33.8	38.3	35.0
Debt service-to-revenue ratio (in percent) 4/	24.5	28.9	34.4	51.9	24.7	32.2	27.2	24.0	22.2	22.0	23.8	25.2	35.4	38.9	36.2
Primary deficit that stabilizes the debt-to-GDP ratio	-3.1	4.3	2.4	1.2	3.8	-5.1	7.0	-2.3	-3.9	-2.1	-1.1	-1.3	1.7	1.2	1.5
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.7	7.3	3.5	5.2	1.2	4.5	20.1	6.8	5.3	5.0	4.8	7.8	4.5	5.7	4.9
Average nominal interest rate on forex debt (in percent)	2.1	3.8	3.3	2.2	0.7	6.2	4.8	5.0	5.2	5.2	5.3	5.3	4.9	5.7	5.2
Average real interest rate on domestic debt (in percent)	-2.4	-3.4	-4.9	-0.9	3.2	3.7	4.4	3.5	4.7	2.5	2.6	3.6	1.7	-0.1	1.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-5.1	9.3	-1.4	-0.4	22.8	-8.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	13.9	16.9	20.7	20.7	7.5	12.8	9.0	7.6	4.2	5.0	5.0	7.3	5.8	7.6	6.1
Growth of real primary spending (deflated by GDP deflator, in percent)	0.2	0.2	-0.1	0.1	0.1	0.1	0.1	0.0	0.1	0.0	0.1	0.1	0.0	0.1	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	13.9	4.2	-0.6	-3.5	-5.4	-6.2	0.4	-8.3	-13.1	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

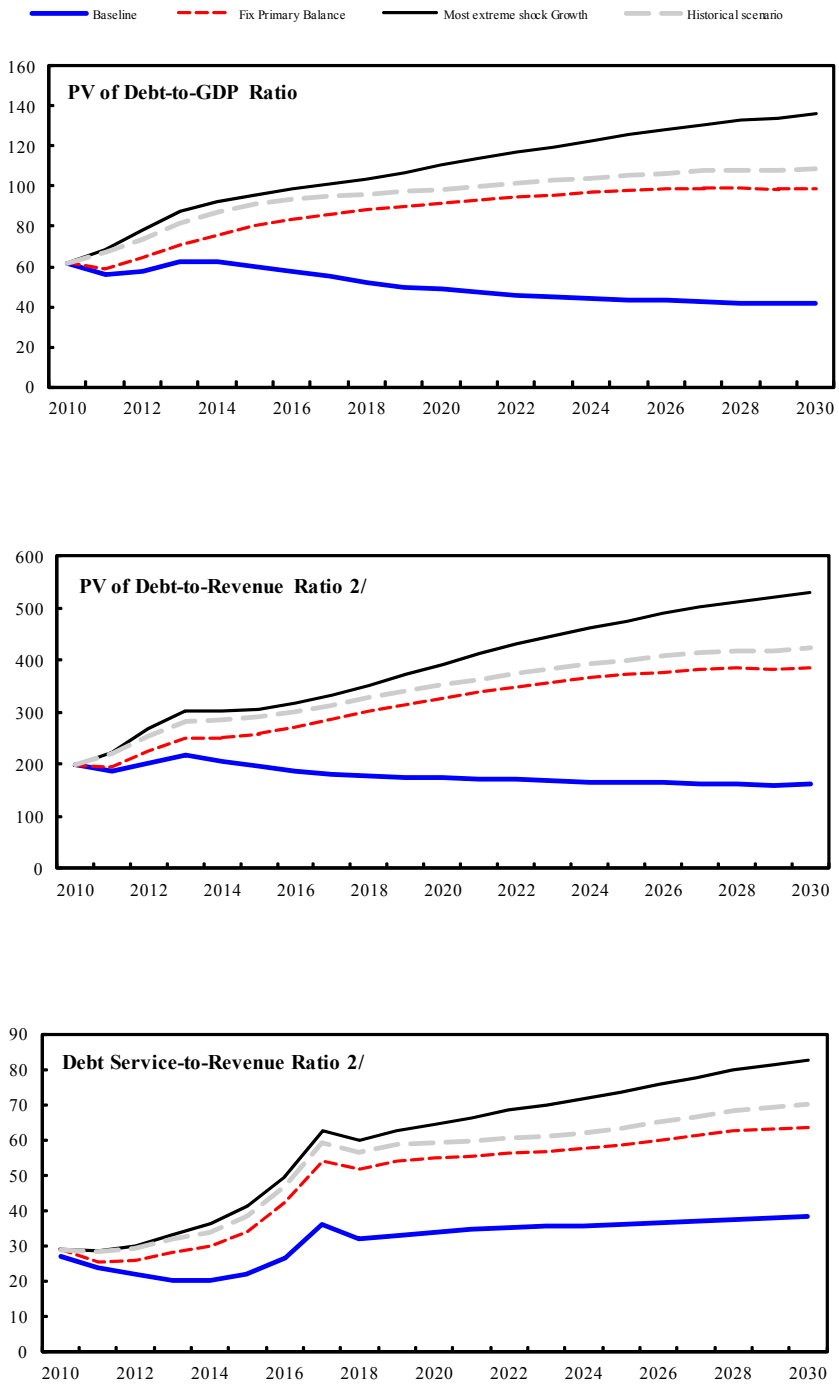
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Figure 2: Indicators of Public Debt Under Alternative Scenarios, 2010-30 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.